

# Towards autonomous imaging workflows using scanning electron microscopy for materials research

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## Abstract

Scanning electron microscopy (SEM) is widely used for materials characterization, but conventional SEM imaging and energy-dispersive X-ray spectroscopy (EDS) acquisition remain dependent on manual navigation, parameter adjustment, region-of-interest selection, and data management. In this work, we develop a semi-autonomous SEM characterization workflow that integrates navigation-camera imaging, automated stage movement, SEM image acquisition, computer-vision-based region selection, targeted EDS analysis, and structured metadata management within a single automated pipeline. Spatial information generated at each stage is used to guide subsequent measurements and maintain traceability among navigation-camera images, SEM images, and EDS spectra. The workflow is demonstrated using two representative classes of materials. In the first case study, multiple circular Cu containing thin-film regions are identified from navigation-camera images and characterized through automated multiscale SEM imaging and spatially registered EDS acquisition. In the second case study, a selected region containing micron-sized Ni particles is systematically scanned. Computer vision is used to detect and filter individual particles, guide targeted center-to-edge EDS measurements, and continue acquisition until 500 particles are characterized. The results demonstrate that the framework can accommodate both pre-defined acquisition patterns and computer-vision-guided selection of features of interest. The proposed workflow enables consistent, traceable, and high-throughput multimodal characterization and supports the future integration of electron microscopy into closed-loop self-driving laboratory and agentic AI workflows.

**Keywords:** self-driving laboratory, automation, scanning electron microscopy, computer vision, energy-dispersive X-ray spectroscopy

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## 1. Introduction

Experimental research consists of three fundamental stages: (i) design of experiments, where experimental conditions are selected to maximize information gain; (ii) execution of experiments, where physical samples or systems are synthesized, processed, and characterized; and (iii) analysis and interpretation of results to extract scientific insight and guide subsequent investigations. In traditional experimentation, these stages rely heavily on human expertise, with researchers selecting experimental conditions, synthesizing or processing materials, characterizing the resulting samples using techniques such as spectroscopy and electron microscopy, and interpreting the acquired data to determine the next experimental steps. Self-driving laboratories (SDLs) [1] aim to automate the workflow and assist human researchers by integrating artificial intelligence, robotics, and autonomous decision-making frameworks. In SDLs, experiment design is often driven by machine learning and optimization algorithms such as Bayesian optimization, which intelligently select the most informative experiments while minimizing

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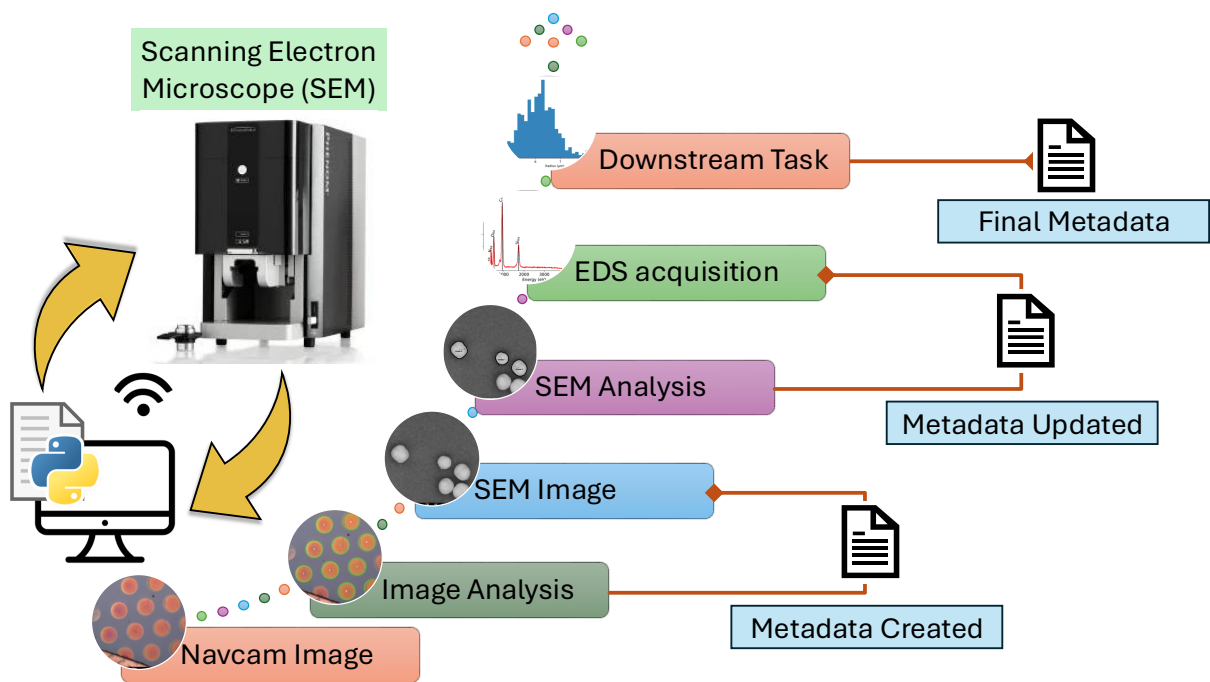


Figure 1: Overview of the semi-automated scanning electron microscopy (SEM) characterization pipeline. The microscope is controlled programmatically through a Python interface. The workflow begins with navigation-camera image acquisition and analysis to identify sample locations, followed by SEM image acquisition and analysis to determine regions for targeted energy dispersive X-ray spectroscopy (EDS) measurements. Acquisition metadata are created and progressively updated at each stage, linking the navigation-camera images, SEM images, EDS spectra, spatial coordinates, and imaging parameters for traceability. Parameters such as horizontal field width and acquisition settings can be provided by the user, whereas information such as sample or feature locations is determined automatically from image analysis. Depending on the characterization task, the user-defined parameters may also be selected automatically. The resulting data and metadata are then used for downstream task.

resource consumption [2, 3]. Experiment execution is automated using robotic platforms [4–6], additive manufacturing systems [7], and other programmable laboratory equipment that operate with minimal human intervention. The resulting samples are subsequently characterized using techniques such as microscopy [8–13], spectroscopy [14], diffraction, mechanical testing, and other characterization techniques, while the acquired data are analyzed using computer vision, machine learning, and data-driven modeling approaches [15–17]. By closing the loop between experimental design, execution, characterization, and analysis, SDLs enable autonomous scientific discovery, accelerate materials development, and significantly reduce the time and cost associated with conventional trial-and-error experimentation.

Among the various characterization techniques, microscopy is widely used to observe structures and features that cannot be resolved by the unaided human eye. It is widely used across disciplines, including medicine, healthcare, biology, materials science, forensics, chemistry, semiconductor manufacturing, environmental science, and nanotechnology [18]. Microscopy techniques can be broadly categorized into optical, electron, scanning probe, and X-ray microscopy. Optical microscopy uses visible light to image specimens at the micrometer scale, while electron microscopy employs electron beams to achieve higher spatial resolution. Scanning probe microscopy utilizes a physical probe to characterize surface topography and local properties at the nanoscale, whereas X-ray microscopy enables non-destructive imaging of internal structures with elemental and chemical contrast.

Among the various microscopy techniques, electron microscopy is widely used when imaging beyond the resolution limits of optical microscopy is required. Electron microscopy can be broadly divided into transmission electron microscopy (TEM) and scanning electron microscopy (SEM). In TEM, a beam of electrons is transmitted through an ultra-thin specimen to generate images and diffraction patterns, enabling the investigation of internal structures and crystallographic information at nanometer to atomic length scales. In contrast, SEM forms images by scanning a focused electron beam across the sample surface and detecting the emitted signals, providing information about surface morphology and topography and, when combined with appropriate detectors, compositional information [19]. Compared to TEM, SEM generally requires less extensive sample preparation, accommodates a wider range of specimen sizes, and offers a larger depth of field, making it a widely used tool for routine characterization and imaging. Furthermore, SEM can be integrated with complementary analytical techniques such as energy-dispersive X-ray spectroscopy (EDS) and electron backscatter diffraction (EBSD), enabling the acquisition of compositional and crystallographic information, respectively.

Despite its widespread use, SEM data acquisition remains highly dependent on human expertise. A typical workflow involves loading samples, navigating to regions of interest, selecting imaging parameters such as magnification, working distance, brightness, and contrast, and then acquiring images and managing data. These tasks are often performed manually and can be time-consuming, limiting experimental throughput and introducing variability between operators. Such manual intervention presents a challenge for the development of fully autonomous SDLs, where experiment design, execution, characterization, and analysis are integrated into a closed-loop framework. Automated SEM operation can address this bottleneck by enabling consistent, high-throughput acquisition of SEM images and associated metadata with minimal human involvement. These data can provide structured inputs for downstream computer vision, machine learning, and decision-making algorithms, thereby accelerating materials characterization and supporting autonomous discovery workflows.

Efforts to reduce this dependence on manual operation have led to the development of increasingly automated electron microscopy workflows (Both TEM and SEM). Early platforms such as Leginon and SerialEM enabled automated target selection, microscope control, tilt-series acquisition, and high-throughput data collection in TEM and cryo-EM [20, 21]. Subsequent work expanded this concept by integrating image analysis, sparse data analytics, and machine learning into the acquisition process, allowing microscope settings and measurement locations to be selected based on information obtained during the experiment [9, 11, 12]. These developments have been especially prominent in TEM and STEM, where experiments require precise instrument alignment, generate information-rich measurements, and often depend strongly on expert operation.

SEM automation has similarly been explored through autofocus and beam optimization, adaptive image acquisition, computer vision-based image analysis, and automated SEM–EDS particle characterization. These approaches reduce operator intervention by automating microscope adjustment, identifying informative regions, extracting quantitative features, and acquiring compositional measurements at selected locations [22–27]. Recent studies have demonstrated image-content-guided data acquisition in commercial SEM systems. Zhang et al. integrated an autonomous SEM module into the CRESSt multimodal robotic platform for electrocatalyst discovery, where automated navigation, focusing, magnification adjustment, image analysis, and microstructural characterization were combined with robotic experimentation and active-learning-based experiment selection [26]. These studies establish that automated SEM operation and its integration into SDLs are feasible. The automation also serves as a necessary prerequisite for developing agentic-AI systems that can interpret characterization data, make measurement decisions, and interact with laboratory equipment with limited human intervention. Building on these advances, the present work develops a semi-autonomous, API-enabled SEM–EDS framework that combines navigation-camera-based sample identification, configurable multiscale imaging, computer-vision-based particle detection, targeted EDS acquisition, and structured metadata management. We demonstrate the workflow using two classes of samples commonly encountered in materials research: thin-film samples and micron-sized particles. The first case study involves automated identification and multiscale imaging of multiple circular copper (Cu) containing thin-film regions, while the second demonstrates systematic tile-based SEM imaging, computer-vision-based detection, and targeted EDS characterization of individual nickel (Ni) particles.

## 2. Methodology

In this work, we have developed a semi-automated multimodal characterization workflow for scanning electron microscopy (SEM). The workflow was implemented on a Thermo Fisher Phenom ProX SEM equipped with a cerium hexaboride ( $\text{CeB}_6$ ) electron source, a backscattered electron (BSE) detector, and energy-dispersive X-ray spectroscopy (EDS) capability. The SEM is also equipped with an optical navigation camera. We utilized the vendor-provided Python programming interface (PPI), a Python-based application programming interface (API), to programmatically control SEM operation.

The PPI enables remote control of the SEM when the control computer and microscope are on the same network. Through this interface, we can switch between navigation camera mode and SEM imaging mode, control stage movement, adjust beam settings, set imaging parameters such as working distance, magnification, brightness, and contrast, and perform EDS acquisition. The ability to modify these parameters through a single Python interface enables building an automated pipeline for multimodal characterization with minimal user intervention.

Our workflow automates SEM characterization by integrating navigation-camera imaging, SEM image acquisition, and EDS spectra collection within a single pipeline. The workflow is shown in Fig. 1. Rather than treating SEM imaging and EDS acquisition as separate manual operations, spatial information generated at each stage is used to guide the subsequent measurement. Navigation-camera images are used to identify sample locations, while SEM images or predefined sampling coordinates are used to determine the positions for targeted EDS acquisition. This spatial linkage ensures that navigation-camera images, SEM images, and EDS spectra remain associated with the same sample region or feature. In addition, the workflow records the corresponding acquisition metadata, allowing each measurement to be traced to its spatial location and imaging conditions.

### 2.1. Navigation-camera imaging

We begin by acquiring a low-magnification overview image using the optical navigation camera (nav-cam). This image provides a wide-field view of the sample holder and is used to define the initial locations for automated characterization. Depending on the sample and the experiment’s objective, the nav-cam image can be used in various ways.

For samples containing multiple discrete specimens, the nav-cam image can be processed directly using computer vision to identify individual samples (samples of interest) and determine their spatial coordinates. These coordinates are then used to move the SEM stage automatically to each specimen for further imaging. This approach is useful when the features of interest are sufficiently large to be resolved in the navigation camera image.

For samples where the features of interest are too small to be identified in the navigation camera image, the navigation image is instead used to define a larger region of interest. The microscope then moves to this region and acquires SEM images, from which the final features of interest are identified. So in some cases, the navigation camera is used for sample-level localization, while in others it is used only to guide the microscope to a region for SEM-based feature detection.

## *2.2. Automated SEM imaging*

After the initial target location or region of interest is defined, the microscope automatically moves to the selected position for SEM imaging. The microscope changes from navigation camera mode to SEM mode. Before image acquisition, the workflow performs the microscope's built-in autofocus routine to determine an appropriate working distance. The working distance, defined as the distance between the sample surface and the objective lens, affects image focus, sharpness, and depth of field. Therefore, selecting an appropriate working distance is essential for acquiring high-quality SEM images. Automatic brightness and contrast adjustment are then applied to adjust image intensity and contrast prior to acquisition.

The SEM imaging strategy depends on the characterization objective. It can be performed at a user-defined fixed location, or the target region of interest can be divided into multiple tiles, with SEM images obtained sequentially. Also, the imaging area, controlled by horizontal field widths (HFWs), can be specified in the framework. Multiple HFWs allow for multi-scale characterization of the region. The number of sampling locations, their radial distance, angular distribution, and HFW values are user-defined parameters and can be modified depending on the sample geometry and characterization objective. For example, the same workflow could be configured to collect images on a grid, along radial or circumferential paths, at randomly sampled positions, or at regions selected using computer vision-based region-of-interest detection.

This flexibility is important for automated SEM characterization because different samples require different acquisition strategies. For relatively uniform samples, fixed-location sampling provides a consistent and repeatable protocol for comparing multiple specimens. For heterogeneous samples, such as those containing particles, defects, cracks, interfaces, or regions with distinct contrast, computer vision algorithms can be incorporated to identify informative regions and guide targeted image acquisition [26]. Depending on the sample, feature localization (sample- or feature-of-interest identification) can be performed either on the navigation camera image or directly on the SEM image, and the detected coordinates are subsequently used for automated characterization. By separating the microscope-control framework from the sampling strategy, the proposed workflow allows users to define simple fixed acquisition patterns or more adaptive ROI-based approaches without modifying the core automation pipeline. This makes the method suitable for high-throughput SEM imaging and provides a pathway for integrating SEM characterization into closed-loop SDL workflows.

During SEM acquisition, the workflow records relevant imaging parameters, including stage coordinates, working distance, accelerating voltage, horizontal field width, brightness, contrast, image file path, etc. These metadata are stored automatically and used to maintain traceability between the acquired SEM images and the later EDS measurements.

## *2.3. Automated EDS acquisition*

Following SEM imaging, the workflow automatically performs EDS acquisition at the selected target locations. These locations may come directly from the navigation-camera-based sample localization step or from feature coordinates identified through SEM image analysis. In both cases, the same

coordinate-based control strategy is used to move the microscope stage and acquire spectra without manual repositioning.

Each EDS spectrum is stored together with its acquisition metadata and linked to the corresponding SEM image. As a result, the final dataset contains navigation-camera images, SEM images, EDS spectra, and structured metadata describing how and where each measurement was collected. This automated multimodal acquisition framework supports high-throughput SEM characterization and possible integration into closed-loop self-driving laboratory workflows.

### 3. Results and discussion

#### 3.1. Case study 1: Semi-automated multiscale characterization of patterned Cu thin-film heterostructures

##### 3.1.1. Thin-film sample preparation

One of the common types of materials that are studied using SEM are thin films to understand their microstructure (grain size, defects), and composition. The thin-film samples used in this study consisted of heterostructures grown by combining atomic layer deposition (ALD) and physical vapor deposition (PVD). A Si substrate with a silicon nitride ( $\text{SiN}_x$ ) layer of 50 nm was used to grow an 8 nm layer of hafnium disulfide ( $\text{HfS}_2$ ) using ALD [28, 29], followed by a top electrode layer of copper (Cu) deposited using a circular shadow mask. Such hetero-structures are being studied to understand barrier layers for copper diffusion in microelectronic applications [30]. The prepared sample contained multiple circular regions that were visible in the SEM nav-cam image. The aim of characterizing these samples is to understand the microstructure of the Cu electrodes grown on  $\text{HfS}_2$  layer as well as composition analysis to understand if the ALD process successfully grew the  $\text{HfS}_2$  layer.

##### 3.1.2. Navigation camera-based sample identification

The navigation camera was used to capture an overview image of the sample holder prior to SEM imaging. The image was acquired at an HFW of 10 mm, as shown in Fig. 2a, providing a wide-field view of the mounted samples. The image is then processed to distinguish between vacant regions, the sample holder, the mounting stubs, and the samples of interest. Since the samples used in this study were circular, we used a Hough circle detection algorithm implemented using OpenCV to identify the samples of interest. The Hough circle method uses the Canny edge detection algorithm to determine the edges. And then uses gradient-based center voting methods to determine all the circles, their corresponding centers, and their radius. This approach is well-suited for detecting circular samples. It does not require model training, labeled datasets, or prior pretraining, unlike supervised computer vision models. Instead, it relies on image processing operations such as grayscale conversion, masking, filtering, edge enhancement, and geometric circle detection. Also, the algorithm is robust to changes in HFWs (which is inversely proportional to magnification).

To improve the accuracy in detecting the Cu samples on the Si substrate, prior knowledge of the sample geometry was incorporated by specifying the expected sample diameter as a user-defined constraint. Specifically, only circles with diameters between 1.6 mm and 1.9 mm were retained, while detections outside this range were discarded. This constraint helped reduce false positives from the sample holder or background and prevented partially visible circular regions from being selected for subsequent SEM imaging, as shown in Fig. 2b. For each localized sample, the circle index, center coordinates, and corresponding diameter were stored in a JSON file, which was subsequently used to guide automated SEM image acquisition.

##### 3.1.3. Automated multi-scale SEM image acquisition

Following the identification of the circular sample regions, the automated SEM image acquisition workflow was executed using the JSON file generated during the localization stage. For each sample, the workflow first moved to the estimated center location and then determined a point on the sample boundary based on the detected diameter. At the sample boundary, the microscope performed the built-in autofocus

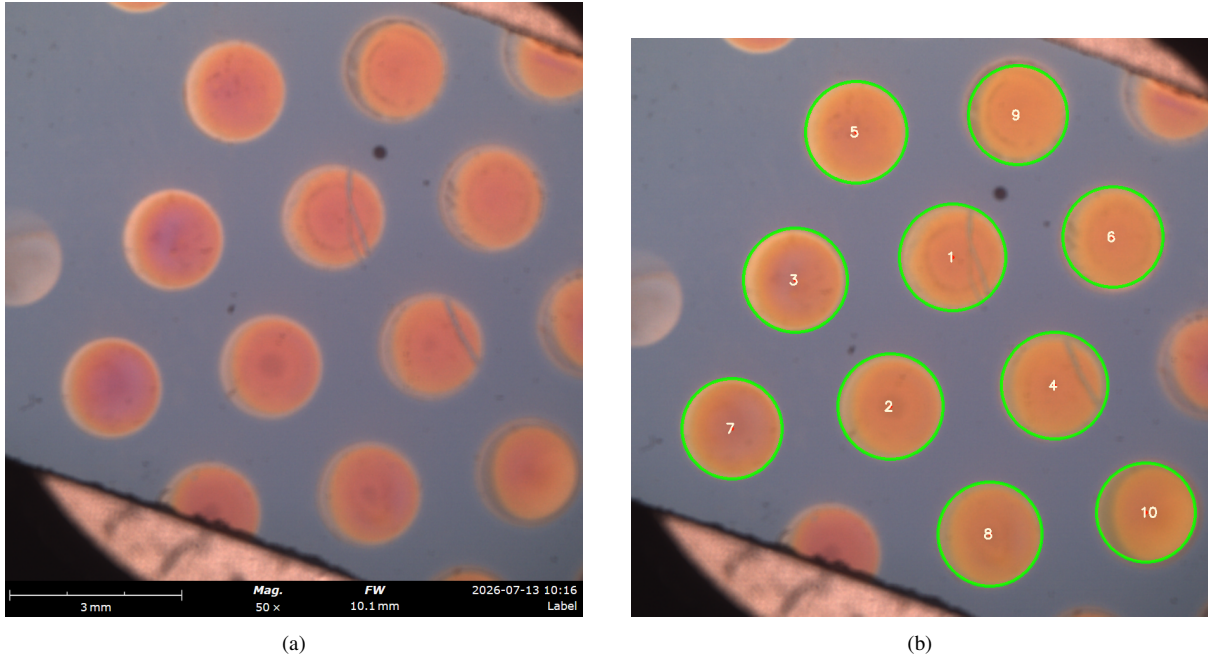


Figure 2: (a) Optical navigation-camera image of the Cu regions deposited on a Si substrate. (b) Circular Cu regions automatically identified using computer vision for subsequent SEM imaging.

and auto brightness/contrast routines at a HFW of 1 mm, which is close to the maximum available HFW in SEM mode (1.2 mm). After optimizing the imaging parameters, the microscope returned to the sample center and automatically acquired SEM images at HFWs of 30  $\mu\text{m}$ , 20  $\mu\text{m}$ , 10  $\mu\text{m}$ , and 5  $\mu\text{m}$ . All SEM images were acquired using the backscattered electron (BSE) detector at an accelerating voltage of 10 kV.

In addition to the sample center, SEM images were acquired at five equally spaced locations positioned at a radial distance equal to 50% of the sample radius. This sampling strategy provided consistent spatial coverage across each circular sample while requiring no user intervention during image acquisition. The SEM images captured at 30  $\mu\text{m}$  HFW for the six locations for circle 1 are shown in Fig. 3. We can see the sample's microstructure is relatively uniform within the entire HFW along the circular pattern. The bright spots seen in the images are due to small contamination accumulated on the sample surface. Although a fixed sampling pattern was used in this study because of the relatively uniform sample geometry, the framework is not restricted to predefined locations. The sampling locations can be modified by the user or determined adaptively using computer vision algorithms to target regions of interest, enabling the same automation framework to support both routine characterization of homogeneous samples and feature-driven characterization of heterogeneous samples. A master metadata containing the image location, HFW, WD, brightness, contrast, accelerating voltage, and associated sample identifier for each SEM image is created. This metadata enables complete traceability of the acquisition process and provides structured information for downstream image analysis, machine learning, and integration into closed-loop SDL workflows.

#### 3.1.4. EDS spectra acquisition

Following SEM image acquisition, the workflow automatically proceeded to collect EDS data, providing an additional characterization modality for elemental analysis. The workflow utilized the master metadata file generated during SEM acquisition to identify the unique sampling locations and subsequently acquired an EDS spectrum at each of these locations. This ensured that every EDS measurement was spatially registered with its corresponding SEM image.

For the experiments presented in this study, the EDS measurements were performed using an accelerating voltage of 5 kV. The accelerating voltage influences the electron interaction volume within the

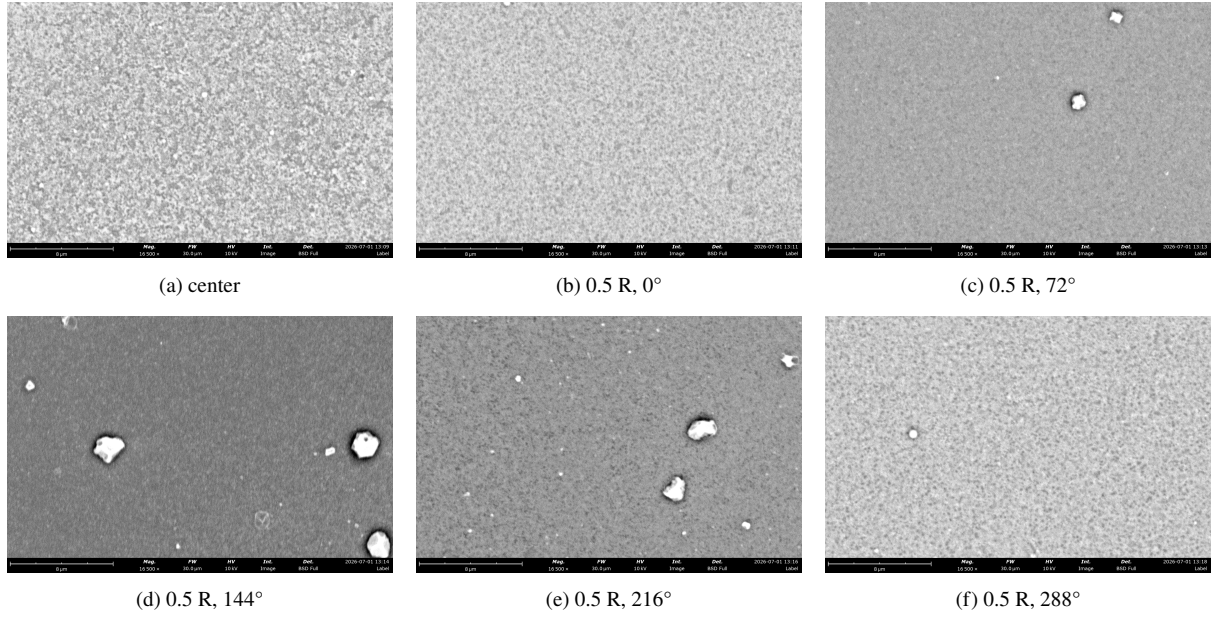


Figure 3: Automated SEM images acquired for Circle 1, shown in Fig. 2b. Images were collected at the sample center and at five equally spaced locations positioned at 50% of the sample radius (R), corresponding to angular positions of 0°, 72°, 144°, 216°, and 288°. All images were acquired at an accelerating voltage of 10 kV using the BSE detector.

sample and, consequently, the depth from which characteristic X-rays are generated. Based on preliminary experiments, an accelerating voltage of 5 kV provided the desired sampling depth for the multilayer samples and enabled the detection of elemental contributions from both Cu and the underlying Si, as shown in Fig. 4. It should be noted that the composition analysis did not show any presence of  $\text{HfS}_2$ . This could be attributed to two reasons - (i) The thickness of the  $\text{HfS}_2$  layer is below the detection limit of the EDS on this SEM, (ii)  $\text{HfS}_2$  was not successfully deposited on the SiN layer using the ALD process.

Such high-throughput characterization of sample's physical structure (patterning), microstructure, and composition analysis can significantly enhance the material synthesis capabilities to enable fine-tuning of growth conditions to obtain desired microstructure and phase.

### 3.2. Case Study 2: Automated SEM–EDS characterization of NMC particles

Nickel-rich layered oxides such as  $\text{LiNi}_x\text{Mn}_y\text{Co}_z\text{O}_2$  ( $x > 0.7$ ) (NMC) are a promising candidate for cathodes in next-generation lithium ion batteries [31]. In particular, using concentration gradient design to vary the Ni content from the core of the particle to its surface has proven to improve the mechanical properties of the cathode particles and reduce susceptibility to cracking while maintaining good electrode capacity. To understand the effectiveness of the synthesis strategies, it is necessary to characterize the size distribution and resulting microstructure of these particles in a high-throughput manner.

The second case study focused on the automated characterization of such Ni-rich NMC particles. The powder consisting of these particles was dispersed on a carbon tape and loaded into the SEM for characterization. Unlike the previous case study, which involved identification of multiple circular samples, this experiment considered a single sample containing numerous NMC particles. Therefore, sample-level localization was not required. A 1.5 mm x 1.5 mm region of interest was selected, and a navigation-camera image of the region was acquired and annotated. The region of interest was scanned sequentially using adjacent SEM image tiles. Each acquired image was processed to identify eligible NMC particles, which were then automatically characterized using EDS. The workflow continued acquiring new SEM images until a total of user-defined 500 particles had been identified, after which the automated acquisition was terminated.

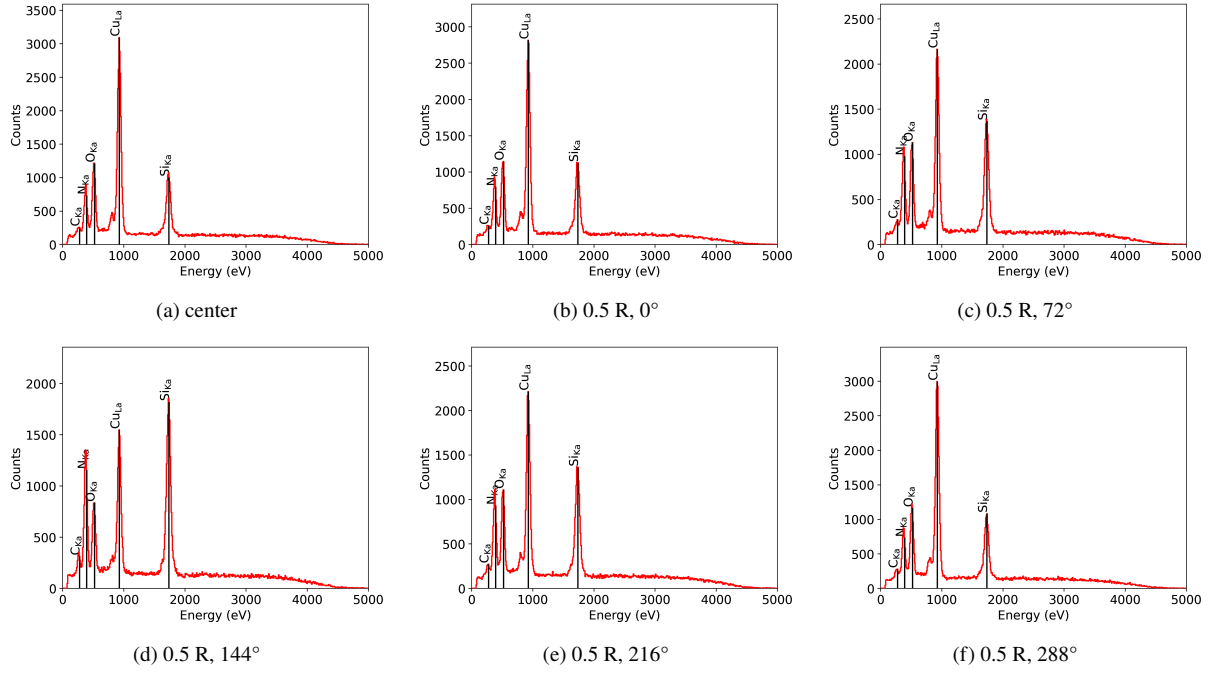


Figure 4: Automated EDS spectra acquired for Circle 10, shown in Fig. 2b. Spectra were collected at the sample center and at five equally spaced locations positioned at 50% of the sample radius (R), corresponding to angular positions of 0°, 72°, 144°, 216°, and 288°. All spectra were acquired at an accelerating voltage of 5 kV.

### 3.2.1. SEM imaging and particle detection

For SEM imaging, an HFW of 73  $\mu\text{m}$  was selected, corresponding to an imaging area of (73 $\mu\text{m}$  x 73 $\mu\text{m}$ ). This HFW was chosen based on preliminary experiments to balance the field of view and image resolution while ensuring that multiple nickel particles were captured in each image. SEM images were acquired using an accelerating voltage of 15 kV and the BSE detector. Prior to image acquisition, the microscope's built-in autofocus and auto brightness/contrast routines were executed to ensure consistent image quality.

Following image acquisition, the SEM images were processed using an automated segmentation pipeline to identify individual nickel particles. The images were first thresholded to separate the particles from the background, after which particle contours were extracted using OpenCV. Particles close to the image boundary were excluded to avoid incomplete measurements resulting from particles being only partially captured within the image. To achieve this, a bounding box was computed for each detected contour, and contours whose bounding boxes intersected or were within a predefined margin of the image boundary were discarded. For each remaining contour, the area ( $A$ ) and perimeter ( $P$ ) were computed using the OpenCV contour analysis functions, and the particle radius was estimated as the mean distance between the contour centroid and all contour points. Particles with an estimated radius smaller than 4.7  $\mu\text{m}$  were excluded to remove smaller particles. Further, a circularity score ( $C$ ) is calculated for each contour by using equation 1.

$$C = \frac{4 * \pi * A}{P^2} \quad (1)$$

Particles above a circularity score of 0.75 was selected for subsequent EDS characterization. The total number of retained particles was simultaneously accumulated using a particle counter, which was later used to terminate the automated workflow after 500 particles had been characterized.

Figure 5a shows a representative SEM image acquired using the automated workflow, while Fig. 5b shows the final particle contours retained after applying the segmentation and filtering criteria. Particles located near the bottom of the image were rejected because the thresholding operation merged adjacent particles into a single connected contour. As a result, the merged contour deviated significantly from a

circular shape, yielding a circularity score below the specified threshold and consequently being excluded.

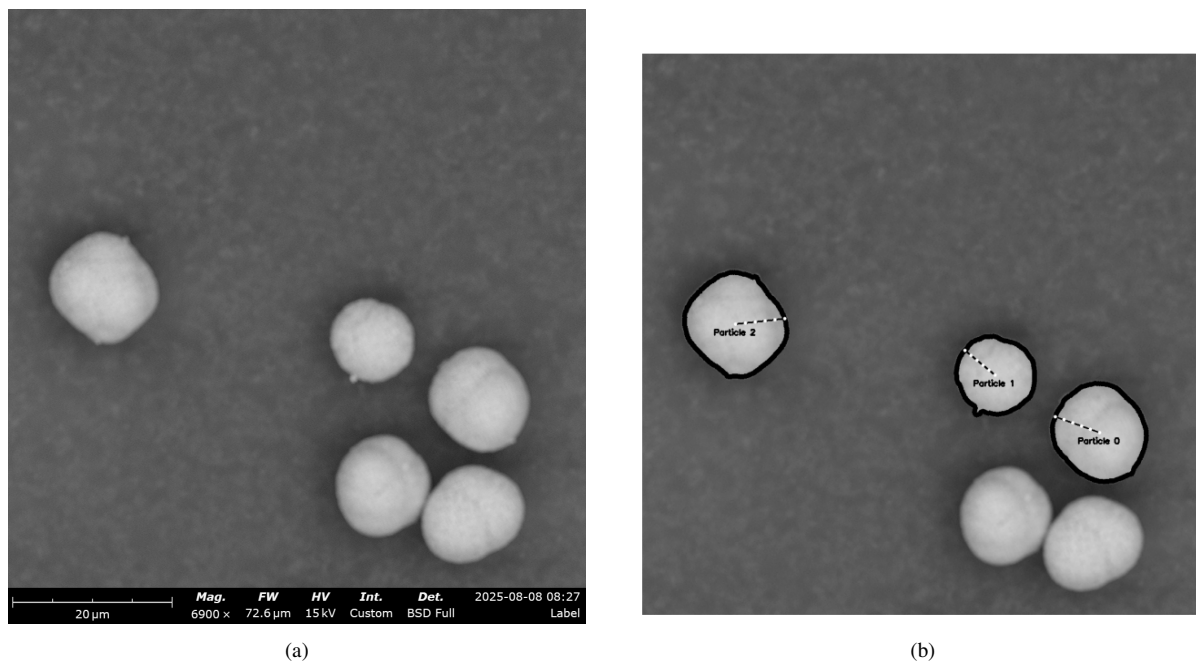


Figure 5: (a) SEM image of the Ni particles (b) Circular particles are automatically identified, and the locations selected for EDS characterization are annotated.

### 3.2.2. Targeted EDS acquisition and particle-size analysis

After identifying an individual NMC particle, an edge point was selected for EDS analysis. The edge point was defined as the contour point whose distance from the particle centroid was closest to the mean particle radius. Using the center and the selected edge point, six equally spaced sampling locations were generated along the line joining the particle center and the edge, as shown in 5b. EDS spectra were then acquired at these six locations using an accelerating voltage of 20 kV, enabling elemental characterization from the particle center toward its boundary. Representative EDS spectra are shown in Fig. [add EDS figure](#).

After completing the EDS measurements at all six sampling locations for a particle, the workflow proceeded to the next detected particle within the same SEM image. The same EDS acquisition procedure was repeated for each particle until all eligible particles in the current SEM image had been characterized. Once all particles in the current image had been analyzed, the stage automatically moved to the next imaging position within the selected 1.5 mm x 1.5 mm region of interest. The region was systematically scanned using adjacent 73 μm x 73 μm SEM images, and the segmentation and EDS acquisition workflow was repeated for each newly acquired image.

A particle counter was maintained throughout the automated characterization process. After each successfully analyzed particle, the counter was incremented, and the workflow continued until a total of 500 particles had been characterized, at which point the acquisition terminated automatically. The particle radius estimated during the segmentation stage was subsequently used to determine the particle size distribution, as shown in Fig. 6. It can be seen that the average particle diameter was determined to be around 6 μm. Although EDX spectra were acquired from the centroid of the particle to the edge, the concentration of Ni is seen to be the same across all measurements. This can be attributed to the fact that the interaction volume of the electron beam with the particle is much smaller than the diameter of the particle, therefore, we only probe the concentration close to the surface. In order to get the true concentration gradient, a cross-section of the particles would be needed.

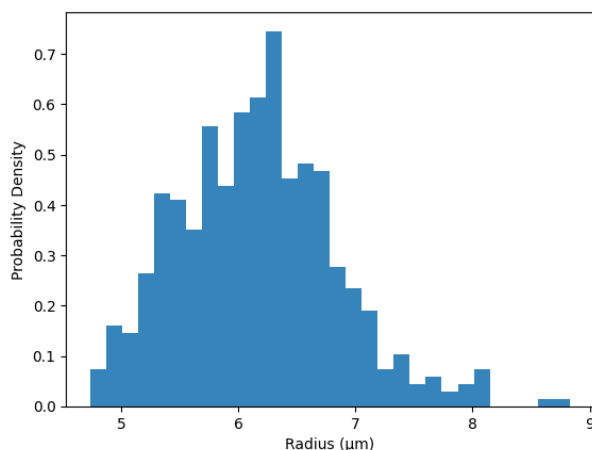


Figure 6: Particle-size distribution of the 500 Ni particles identified by the automated workflow, demonstrating particle-size analysis as a downstream task.

## 4. Conclusion

We presented a semi-autonomous, API-enabled workflow for multimodal SEM–EDS characterization that bridges the gap between traditional, operator-driven microscopy and the requirements of self-driving laboratories. By leveraging vendor-provided Python APIs, navigation-camera imaging, configurable multiscale SEM acquisition, computer-vision-based feature detection, and targeted EDS measurements within a single, coordinated pipeline, the framework demonstrates how SEM can be transformed from a manual characterization tool into a high-throughput, programmable component of autonomous materials research workflows. The two case studies, Cu thin films with predefined electrode regions and heterogeneous NMC cathode particles, demonstrate that the same architecture can support both region-centric and feature-centric characterization, delivering spatially resolved microstructural and compositional data suitable for downstream analysis.

Looking forward, the current semi-autonomous workflow can be extended in several directions. On the microscopy side, additional modalities can be integrated. On the algorithmic side, more advanced computer vision and machine learning models could replace fixed sampling rules with adaptive, information-driven measurement strategies that prioritize defects, interfaces, or rare microstructural features. Because regions of interest vary across materials and characterization objectives, their automated identification will remain application dependent. And at the system level, coupling the workflow to experiment planners and agentic AI controllers would enable fully autonomous discovery loops, in which SEM–EDS data directly inform subsequent synthesis or processing steps. By demonstrating a working implementation on a commercial SEM platform, this study offers a practical step toward these future capabilities and highlights the important role of automated microscopy in self-driving laboratories for materials research.

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## Competing interests

The authors declare no competing interests.

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